

2-Mark Questions & Answers -- Data Structures Module 5: Graphs

Q1. Define a Graph and state the Handshaking Lemma.

A Graph is $G = (V, E)$ consisting of a set of vertices V and edges E .

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Q2. Differentiate between Directed and Undirected graphs.

- * Undirected Graph: Edges are bidirectional and symmetric $((u, v) = (v, u))$.

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Q3. What is Adjacency Matrix vs Adjacency List

- * Adjacency Matrix: 2D array of size $V \times V$, $O(V^2)$ space, $O(1)$ edge query.

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Q4. Define Topological Sorting.

Topological Sorting of a Directed Acyclic Graph (DAG) is a linear ordering of vertices such that for every directed edge (u, v) , vertex u appears before v in the ordering.